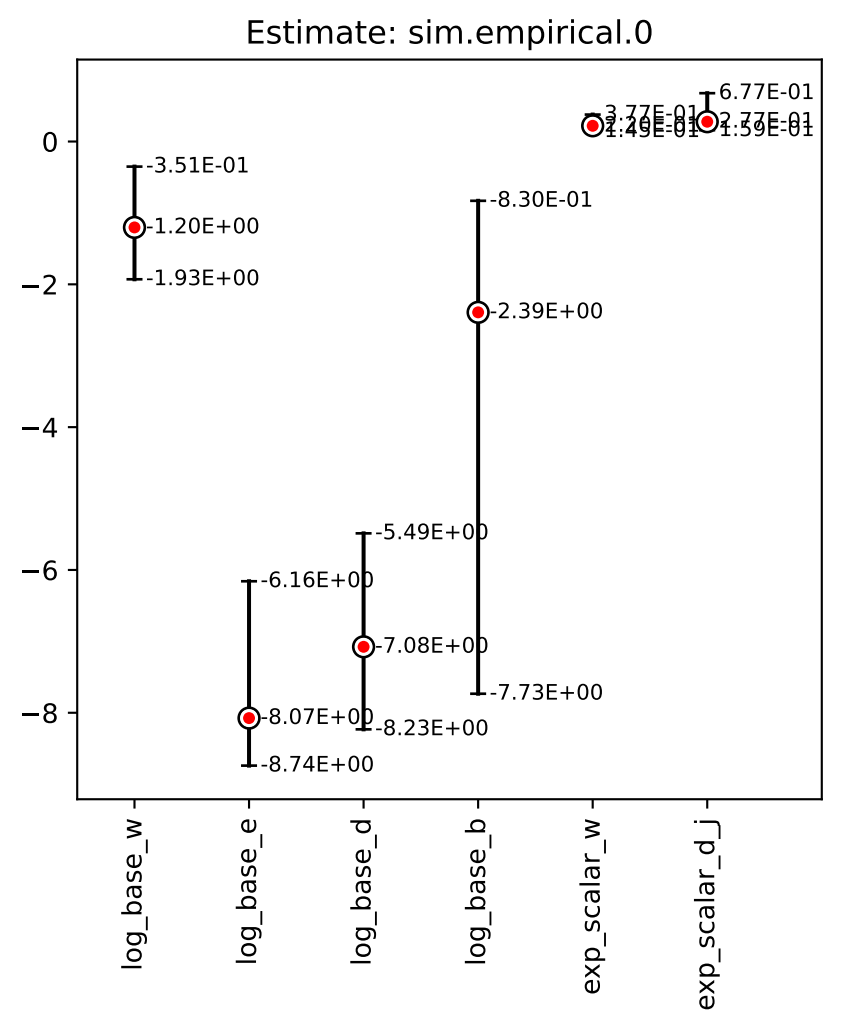
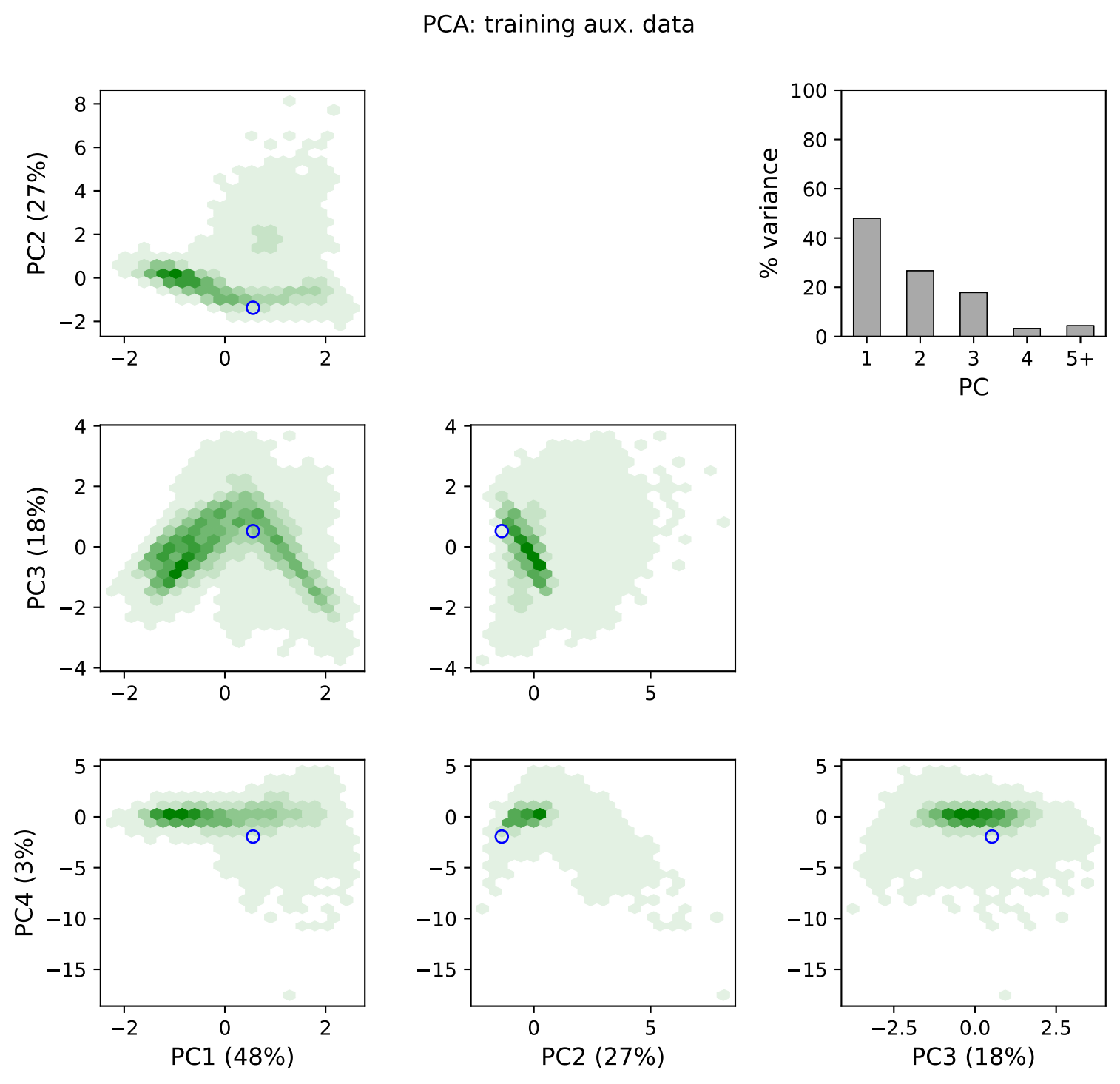
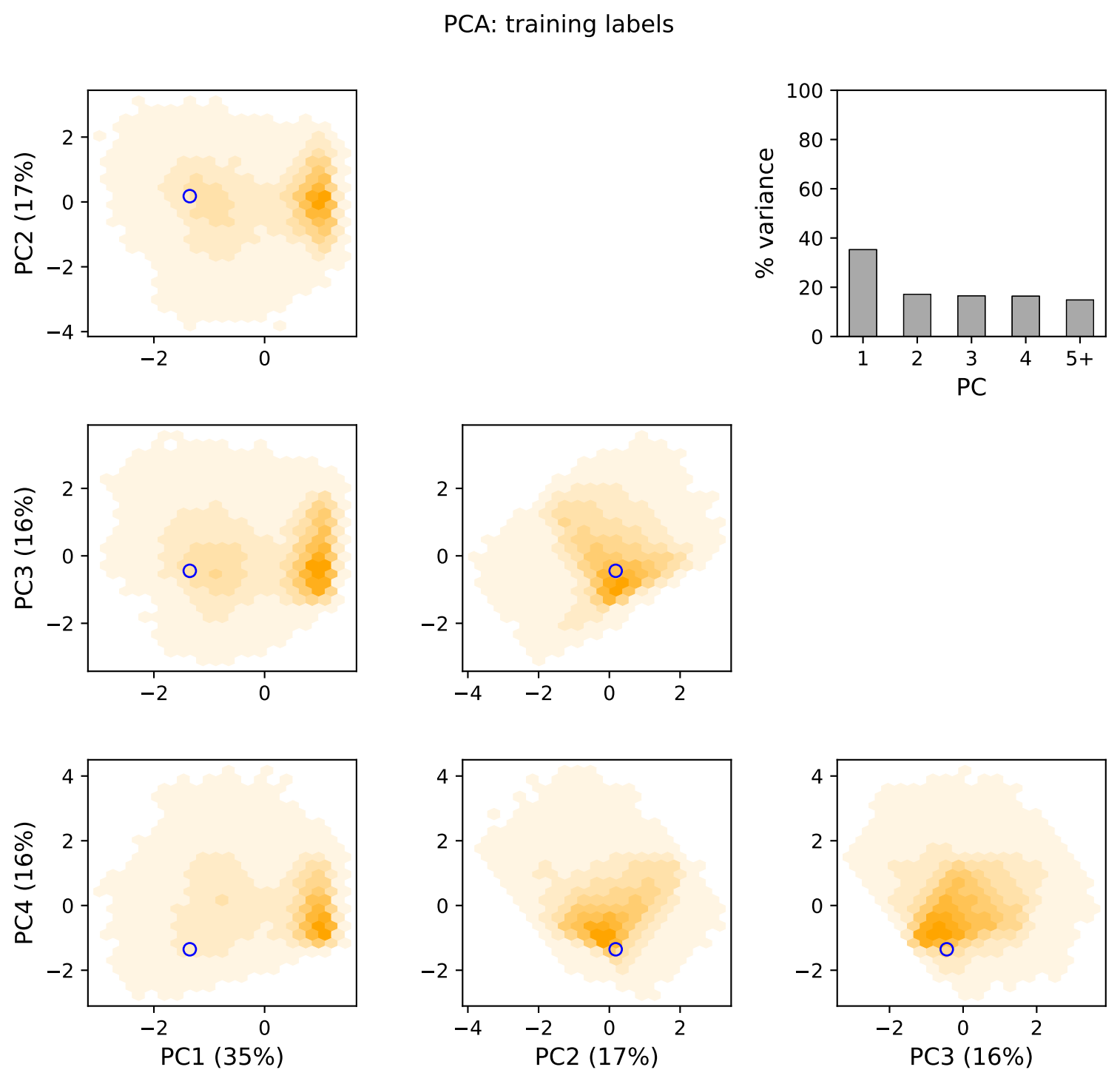
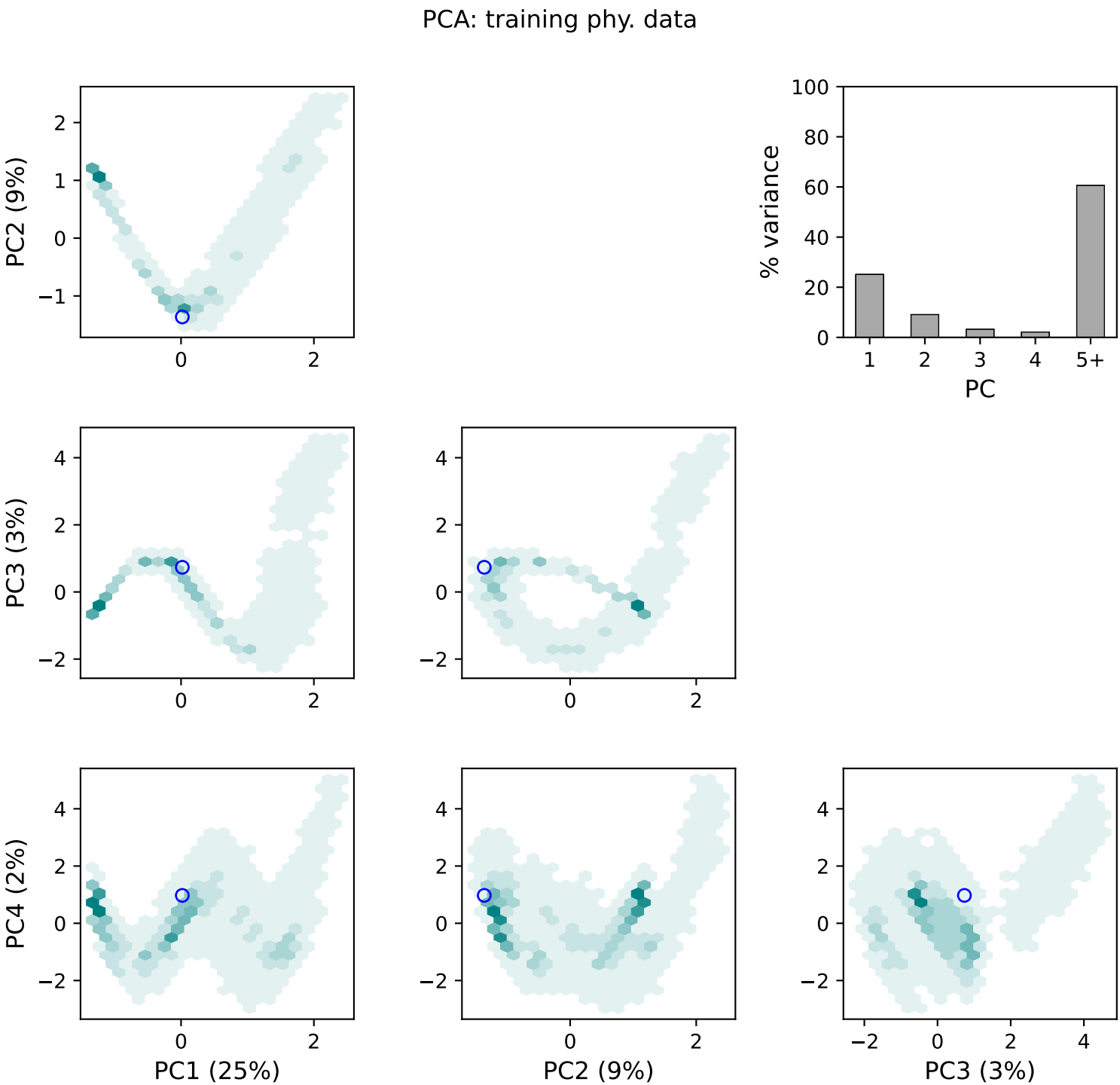
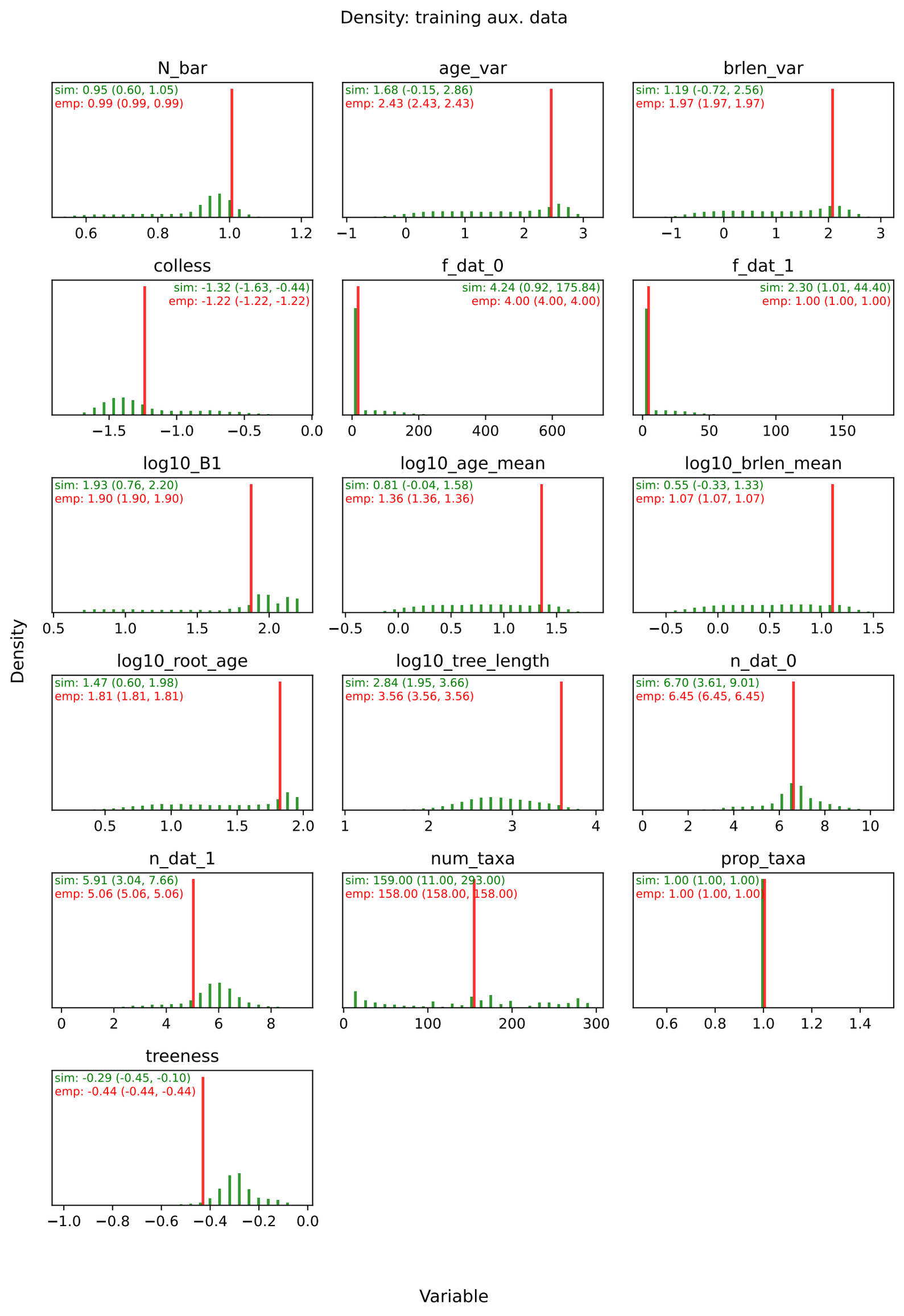
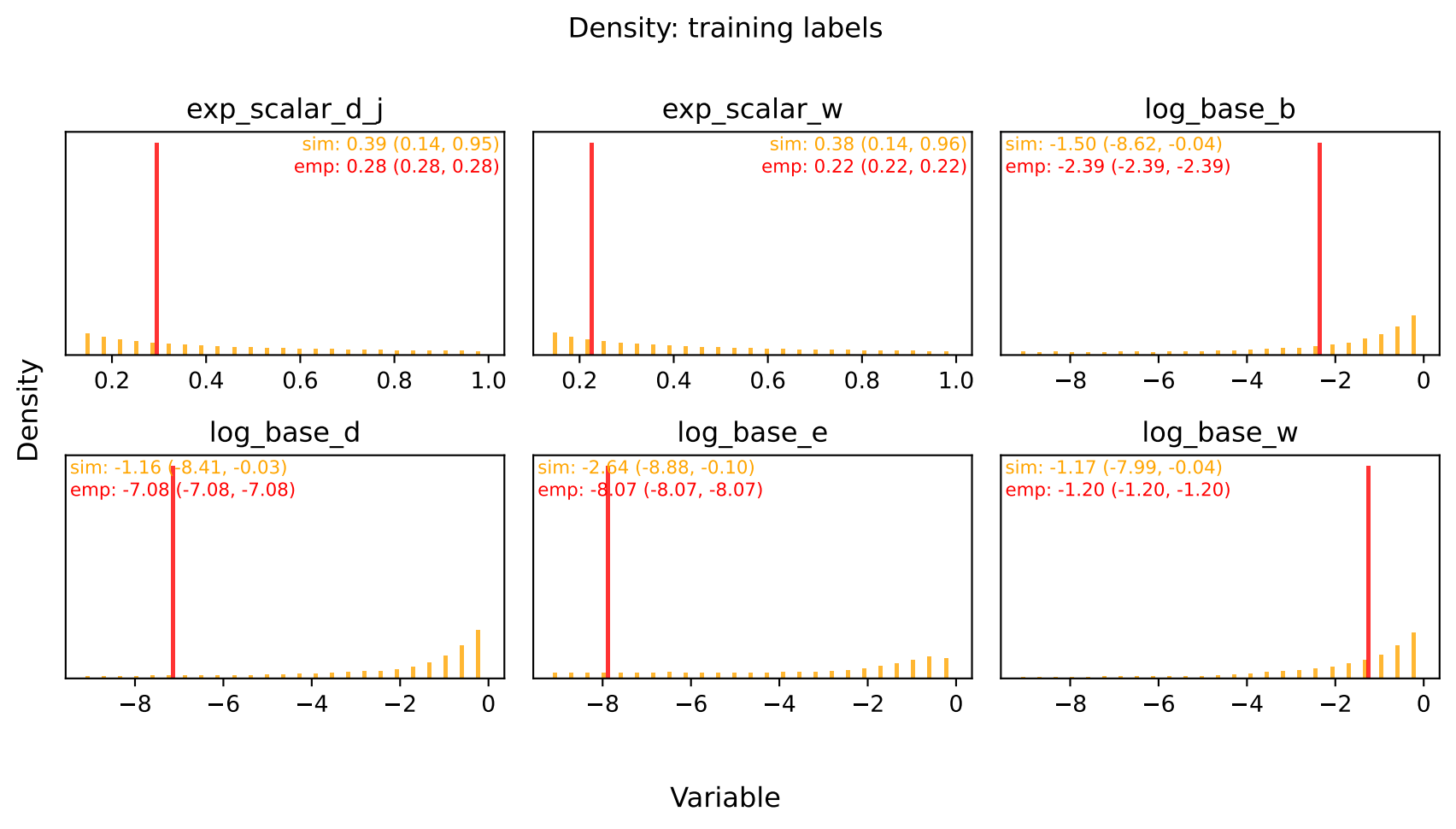

Estimate: sim.empirical.0
0
−2
−4
−6
−8
-3.51E-01
-1.20E+00
-1.93E+00
-6.16E+00
-8.07E+00
-8.74E+00
-5.49E+00
-7.08E+00
-8.23E+00
-8.30E-01
-2.39E+00
-7.73E+00
6.77E-01
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_d_j

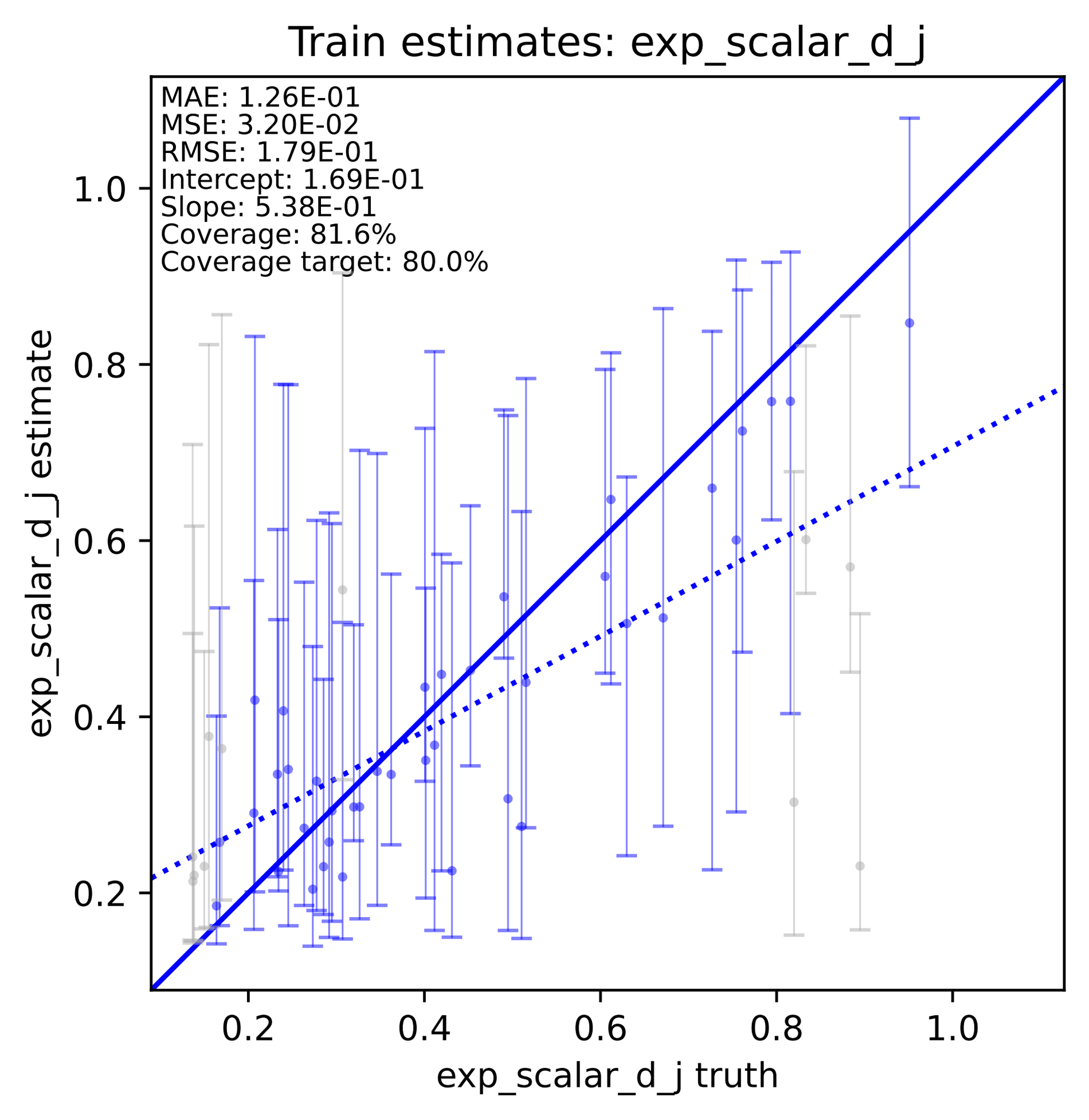

PCA: training aux. data
PC2 (27%)
% variance
PC
PC3 (18%)
PC4 (3%)
PC1 (48%)
PC2 (27%)
PC3 (18%)

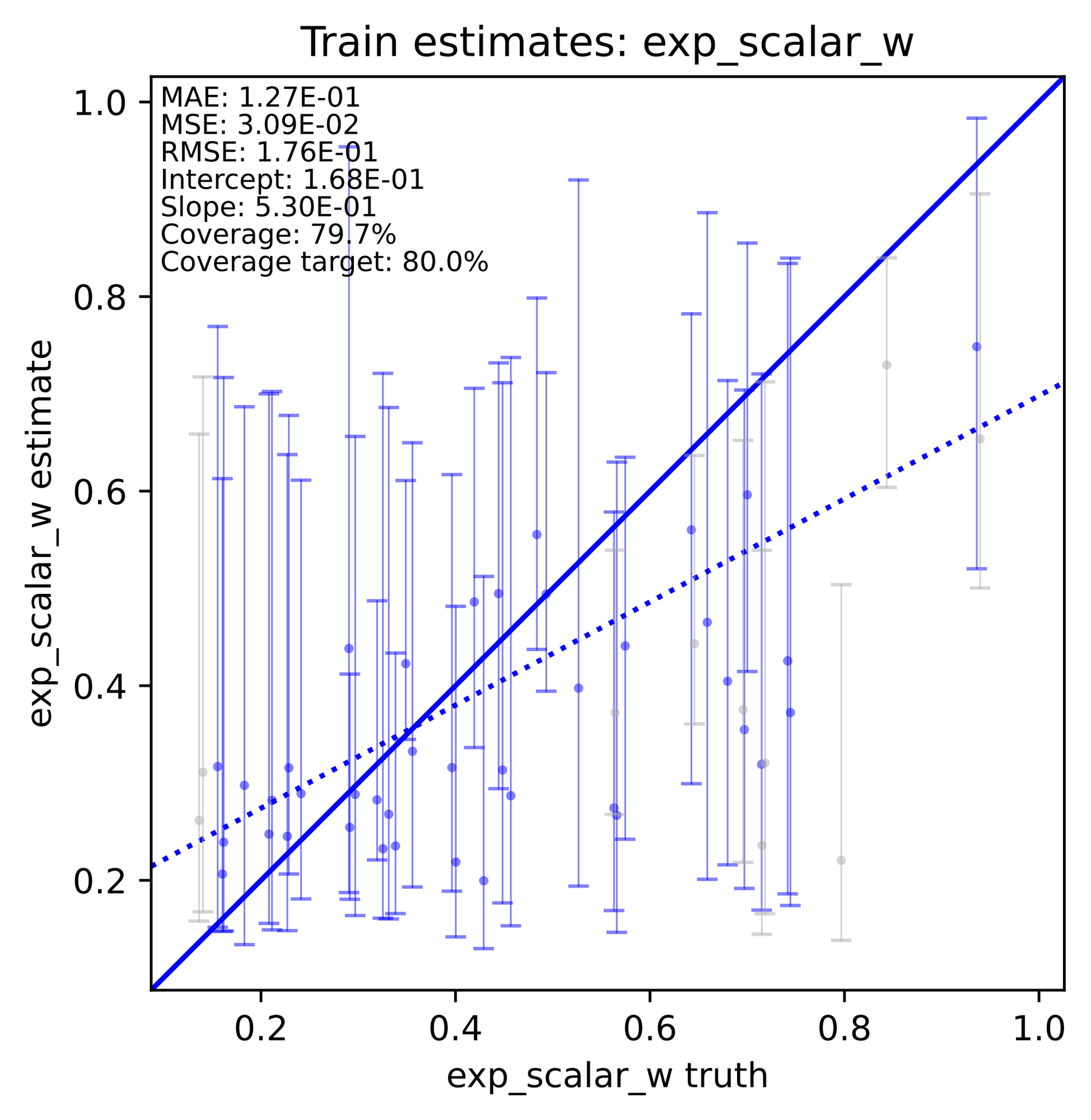

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (16%)
PC4 (16%)
PC1 (35%)
PC2 (17%)
PC3 (16%)

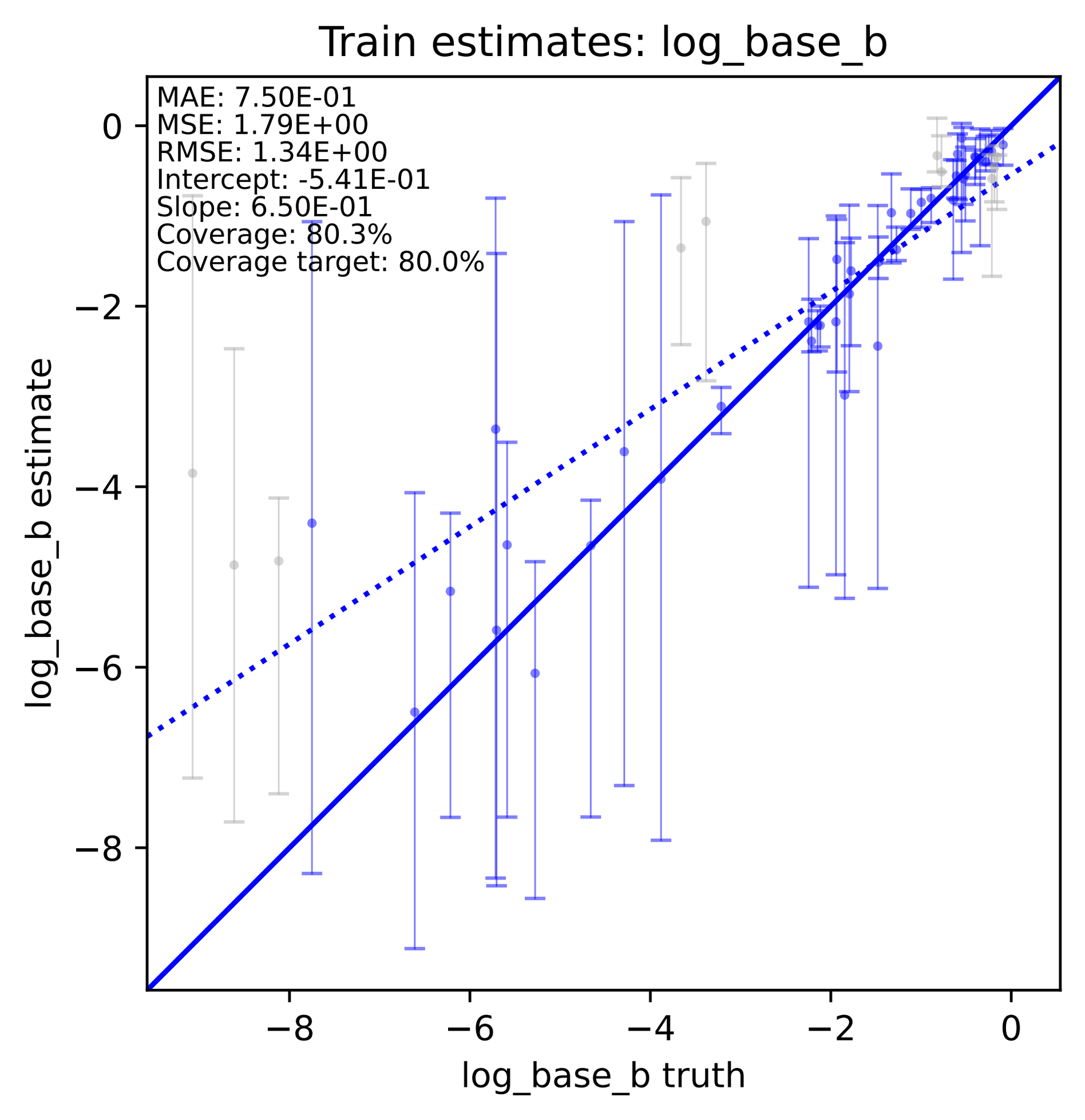

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (3%)
PC4 (2%)
PC1 (25%)
PC2 (9%)
PC3 (3%)

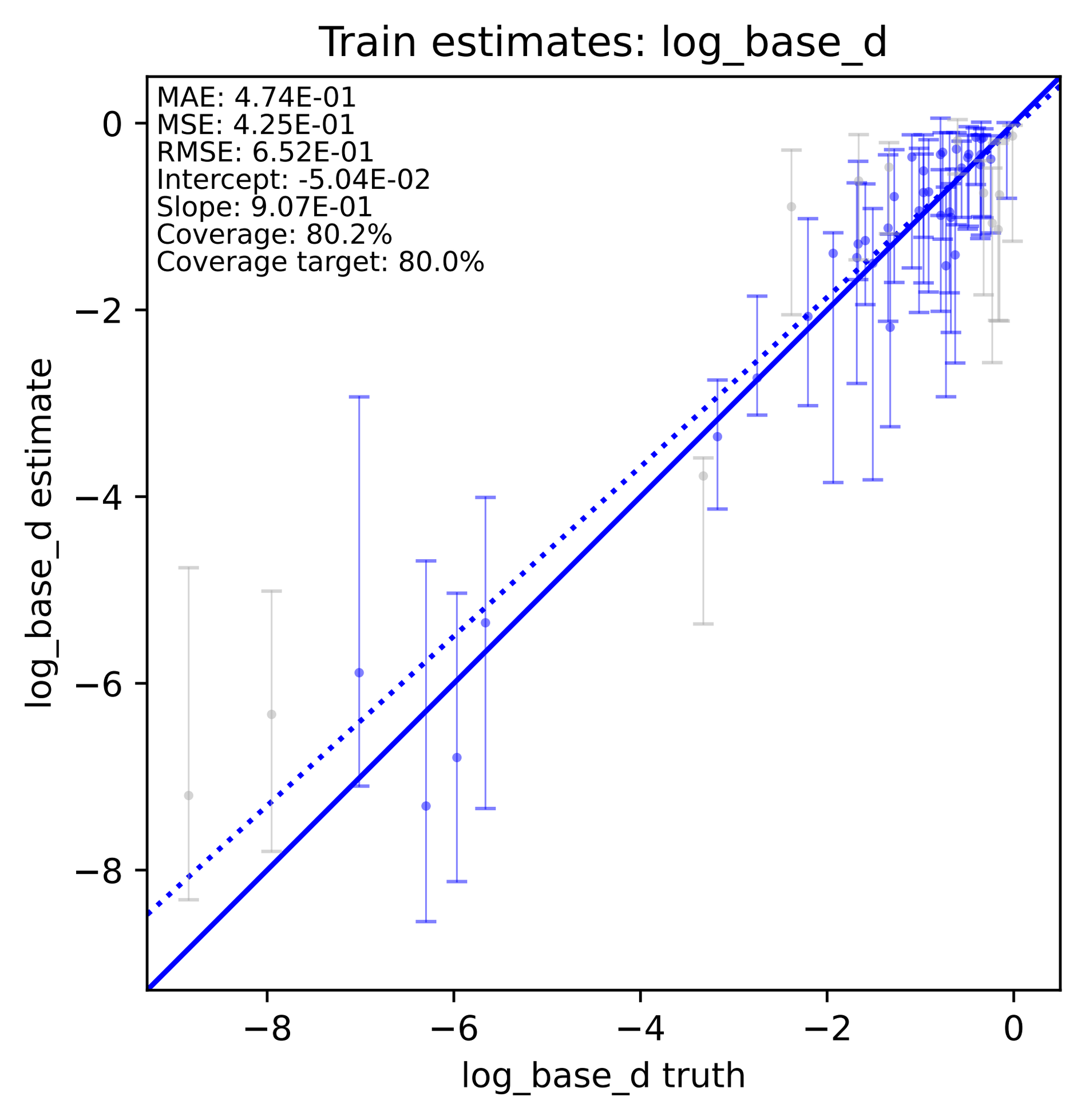

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.05)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.68 (-0.15, 2.86)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 1.19 (-0.72, 2.56)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.32 (-1.63, -0.44)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.24 (0.92, 175.84)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.30 (1.01, 44.40)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.93 (0.76, 2.20)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.81 (-0.04, 1.58)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.55 (-0.33, 1.33)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.47 (0.60, 1.98)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.84 (1.95, 3.66)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.70 (3.61, 9.01)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.91 (3.04, 7.66)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 159.00 (11.00, 293.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.29 (-0.45, -0.10)
emp: -0.44 (-0.44, -0.44)
Density
Variable

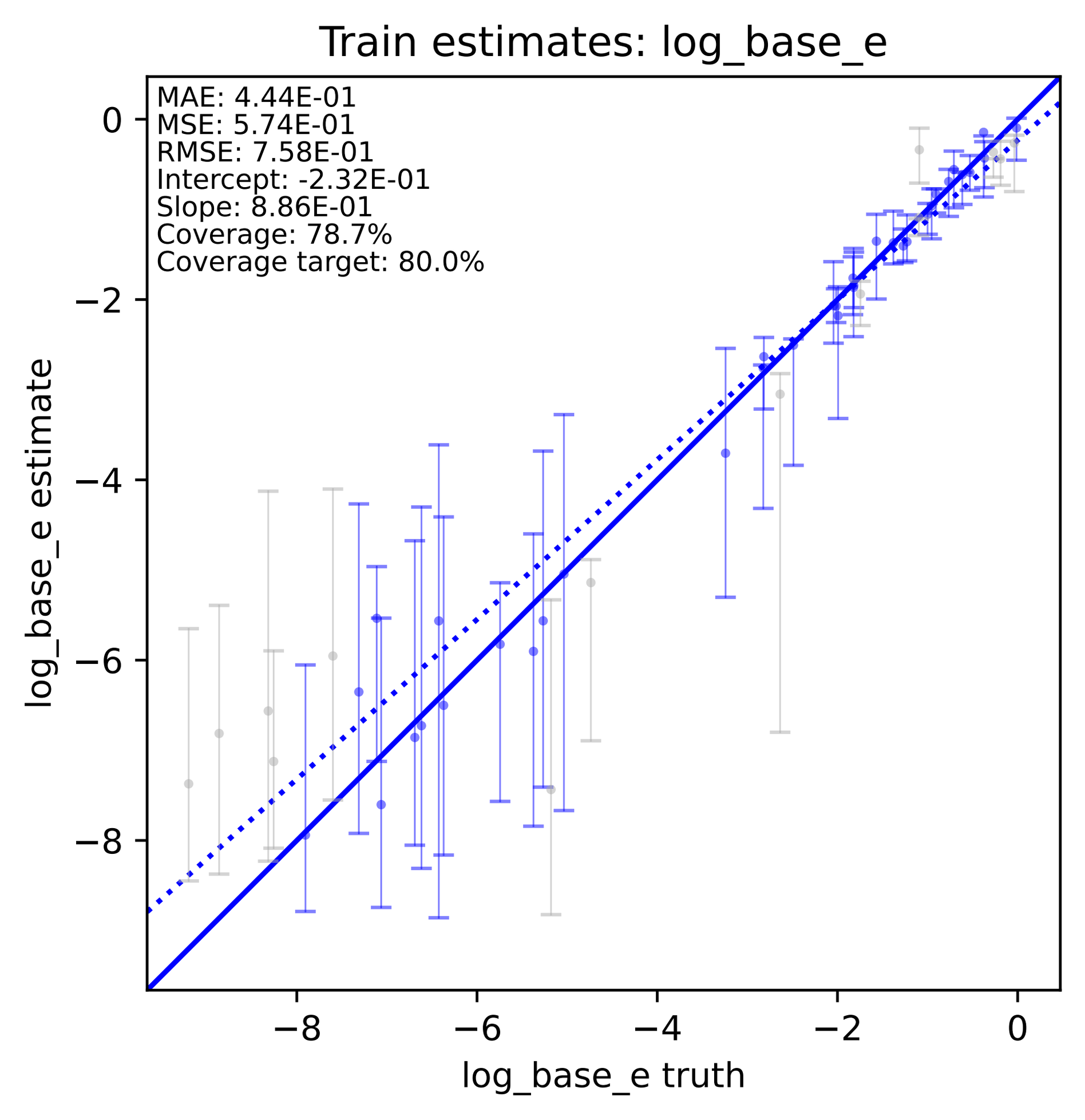

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.95)
emp: 0.28 (0.28, 0.28)
exp_scalar_w
sim: 0.38 (0.14, 0.96)
emp: 0.22 (0.22, 0.22)
log_base_b
sim: -1.50 (-8.62, -0.04)
emp: -2.39 (-2.39, -2.39)
log_base_d
sim: -1.16 (-8.41, -0.03)
emp: -7.08 (-7.08, -7.08)
log_base_e
sim: -2.64 (-8.88, -0.10)
emp: -8.07 (-8.07, -8.07)
log_base_w
sim: -1.17 (-7.99, -0.04)
emp: -1.20 (-1.20, -1.20)
Density
Variable

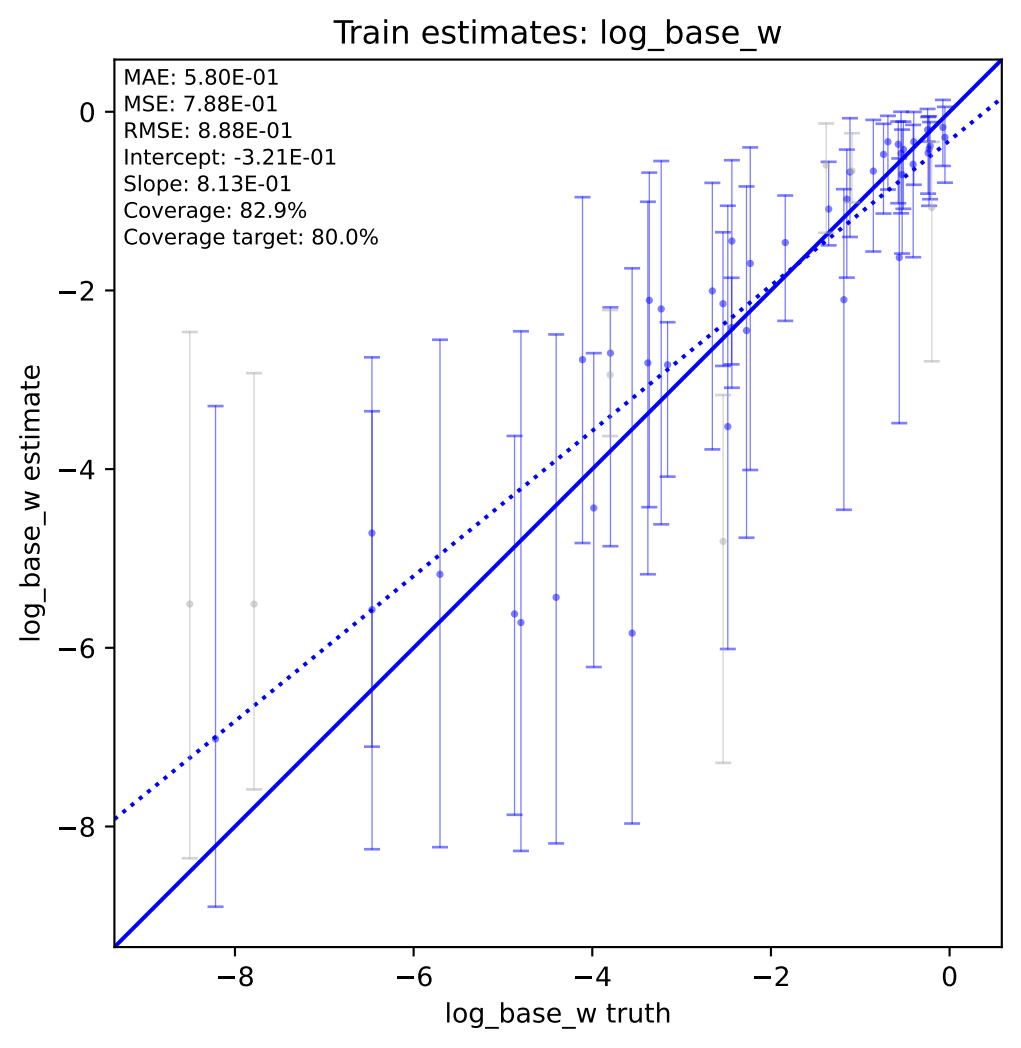

Train estimates: exp_scalar_d_j
MAE: 1.26E-01
MSE: 3.20E-02
RMSE: 1.79E-01
Intercept: 1.69E-01
Slope: 5.38E-01
Coverage: 81.6%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

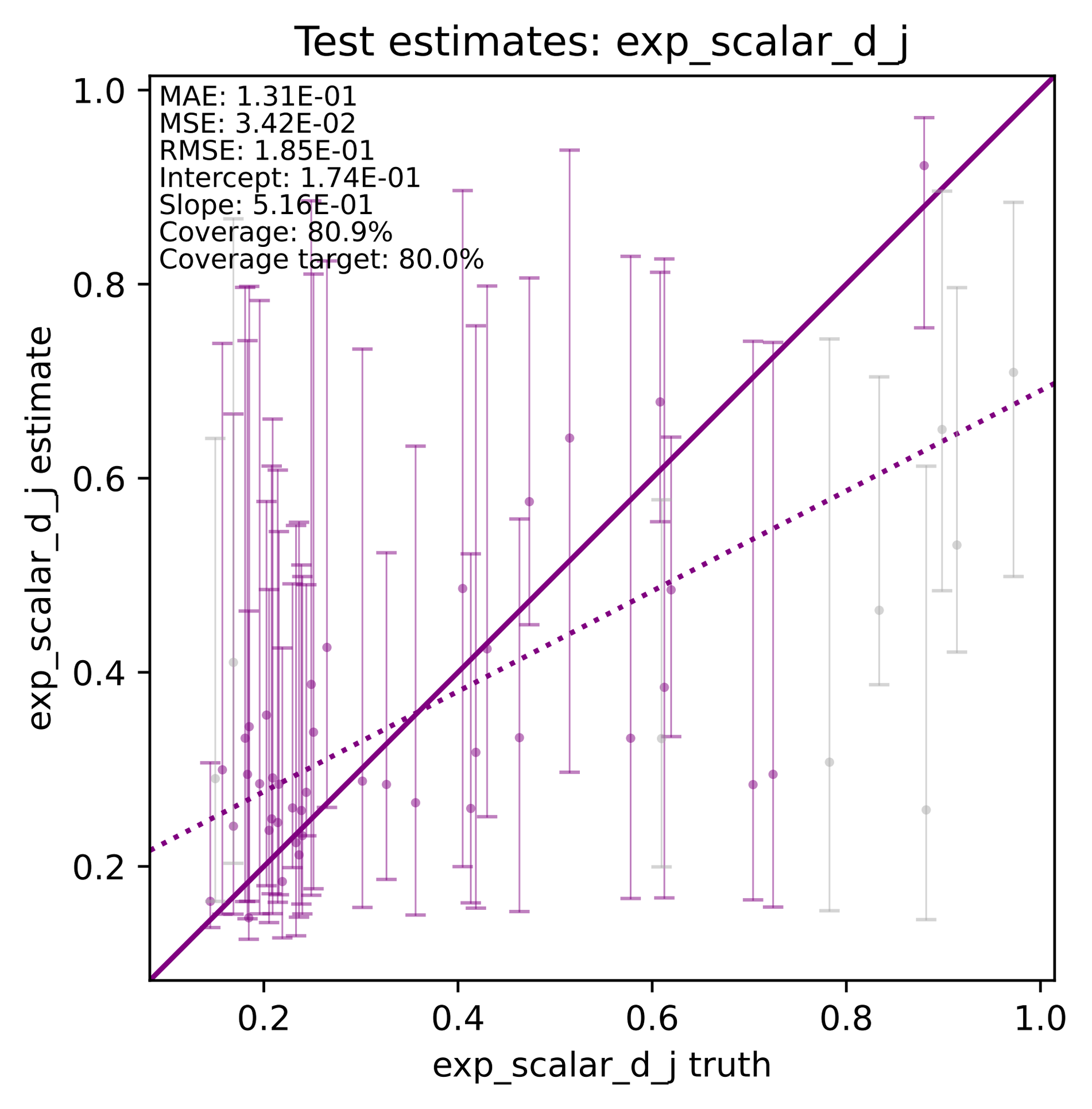

Train estimates: exp_scalar_w
MAE: 1.27E-01
MSE: 3.09E-02
RMSE: 1.76E-01
Intercept: 1.68E-01
Slope: 5.30E-01
Coverage: 79.7%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

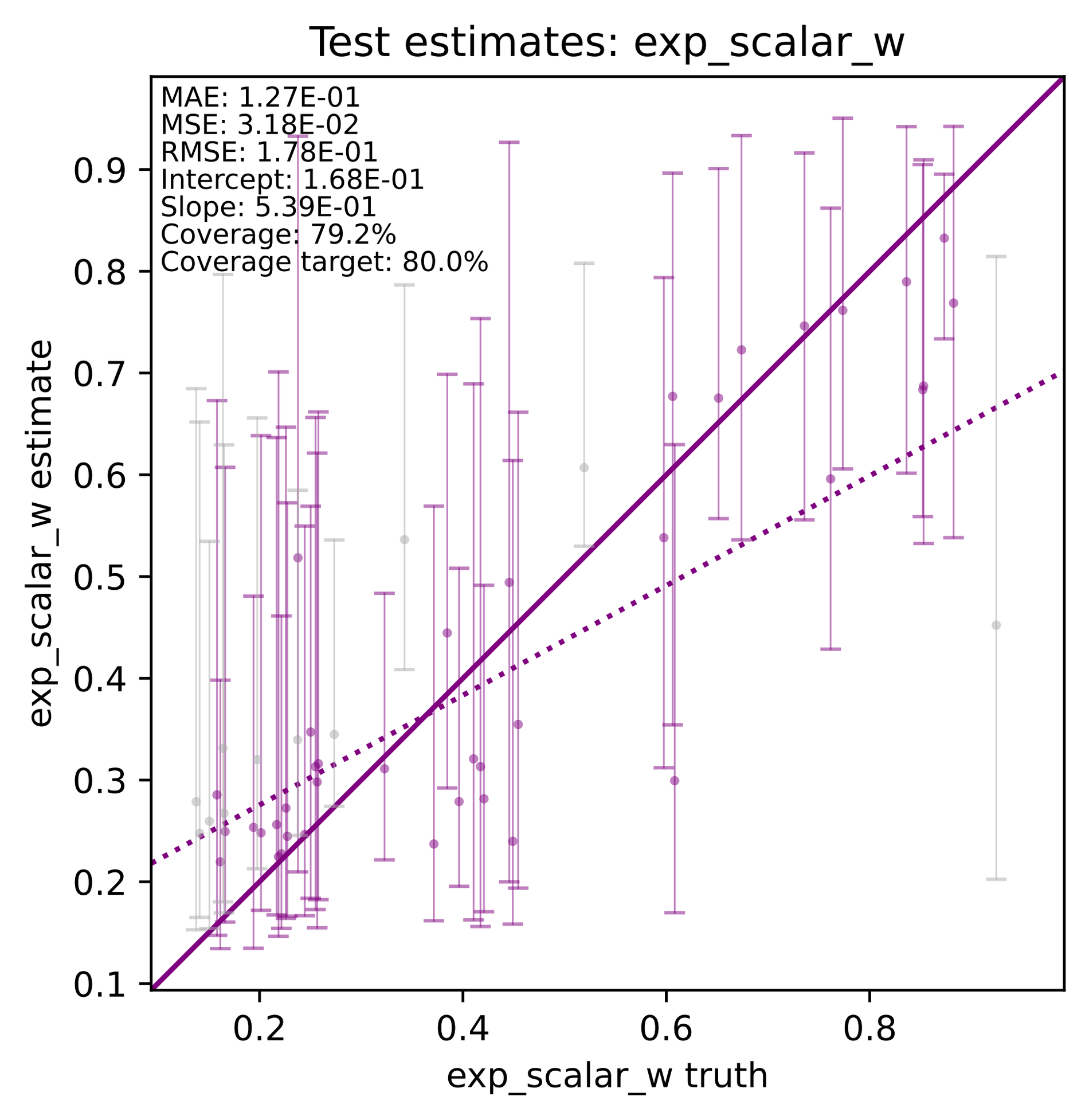

Train estimates: log_base_b
MAE: 7.50E-01
MSE: 1.79E+00
RMSE: 1.34E+00
Intercept: -5.41E-01
Slope: 6.50E-01
Coverage: 80.3%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

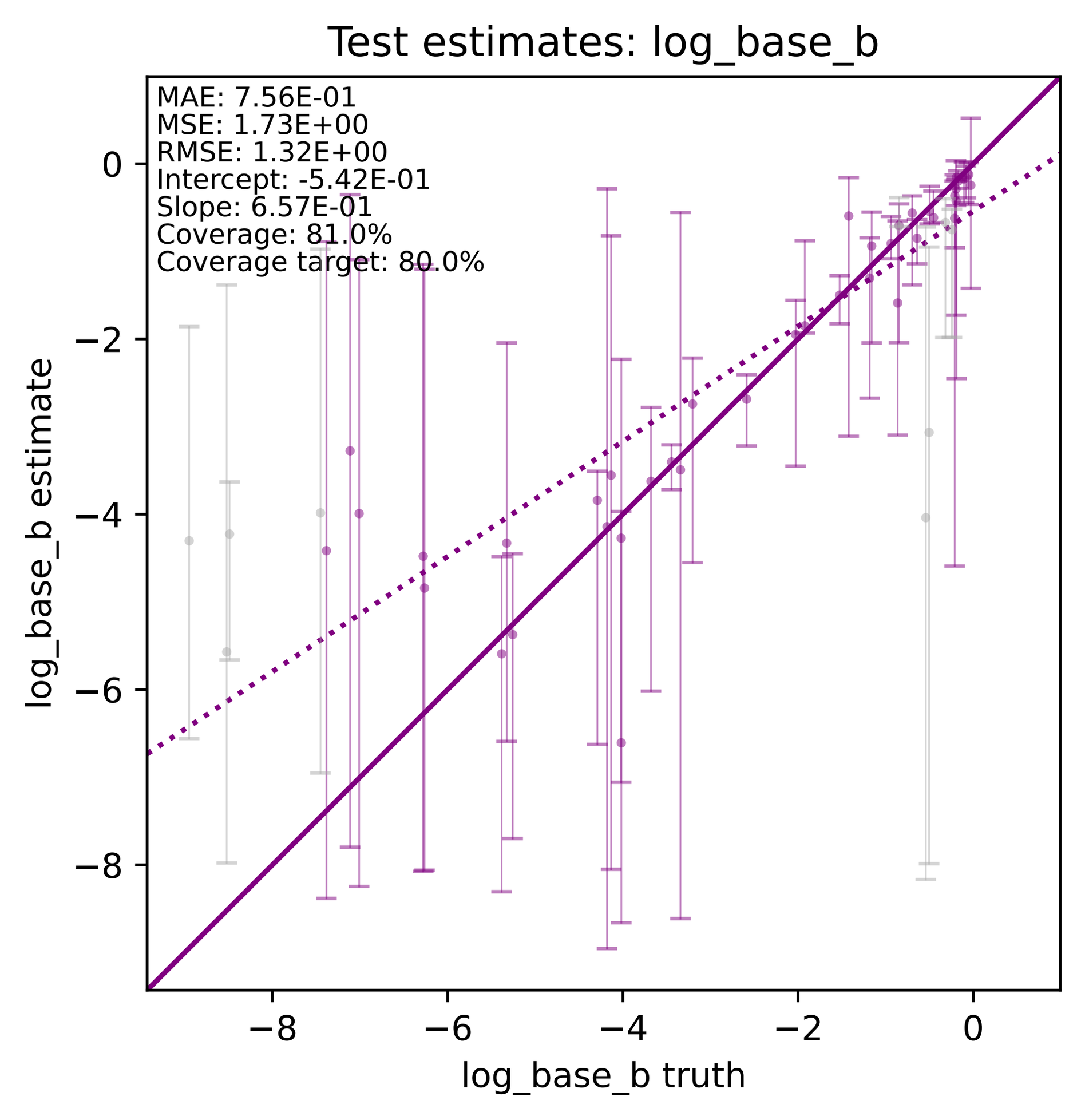

Train estimates: log_base_d
MAE: 4.74E-01
MSE: 4.25E-01
RMSE: 6.52E-01
Intercept: -5.04E-02
Slope: 9.07E-01
Coverage: 80.2%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

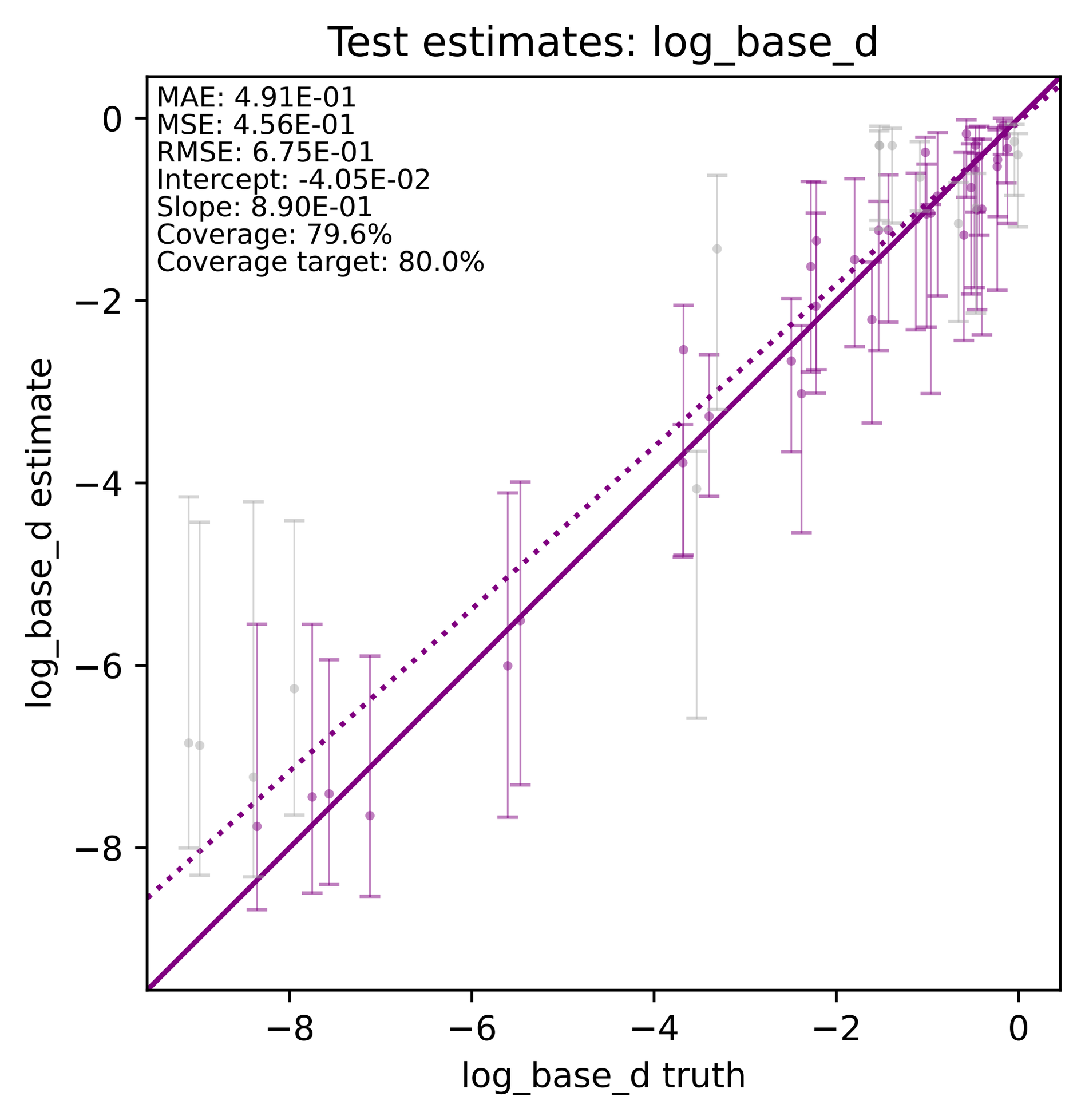

Train estimates: log_base_e
MAE: 4.44E-01
MSE: 5.74E-01
RMSE: 7.58E-01
Intercept: -2.32E-01
Slope: 8.86E-01
Coverage: 78.7%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

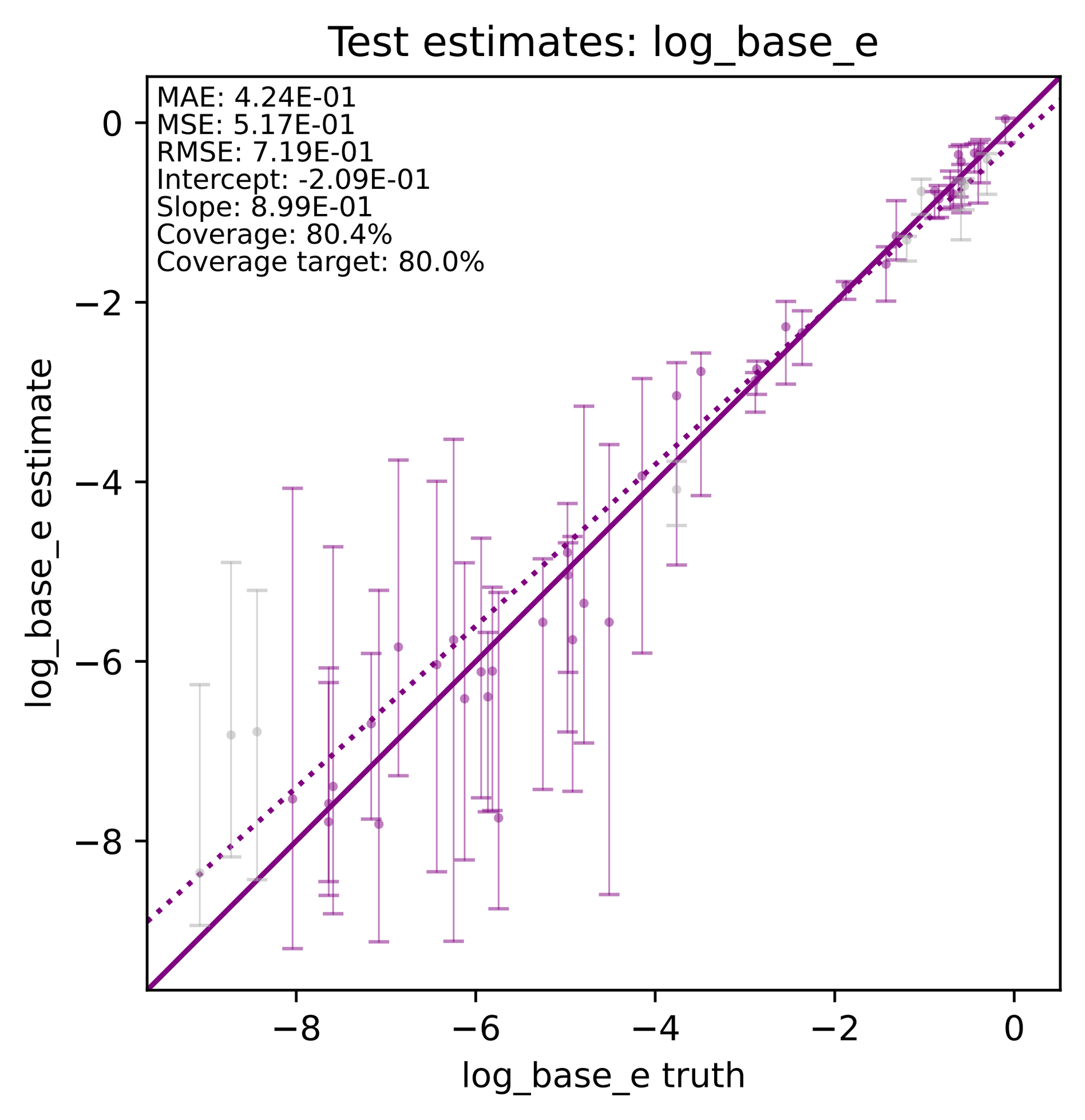

Train estimates: log_base_w
MAE: 5.80E-01
MSE: 7.88E-01
RMSE: 8.88E-01
Intercept: -3.21E-01
Slope: 8.13E-01
Coverage: 82.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

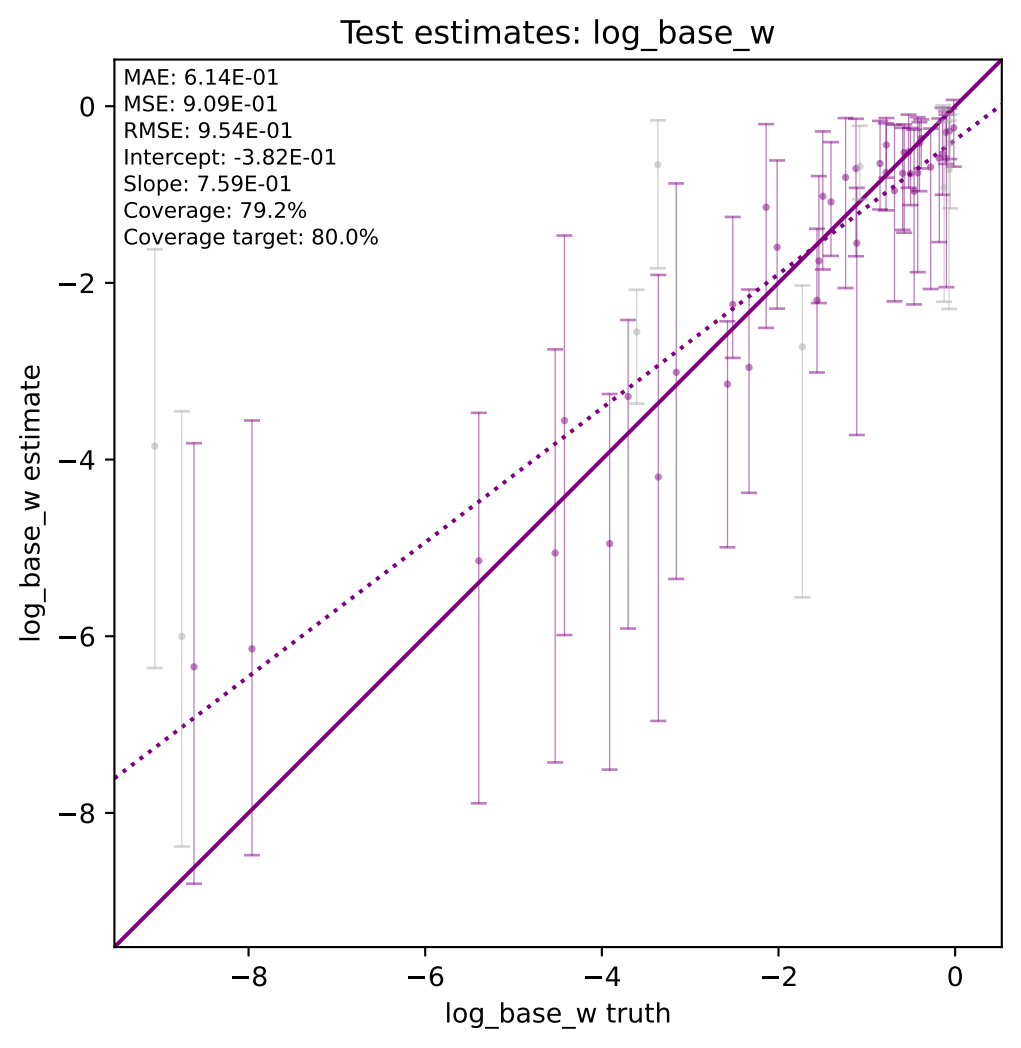

Test estimates: exp_scalar_d_j
MAE: 1.31E-01
MSE: 3.42E-02
RMSE: 1.85E-01
Intercept: 1.74E-01
Slope: 5.16E-01
Coverage: 80.9%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
1.0
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1.0

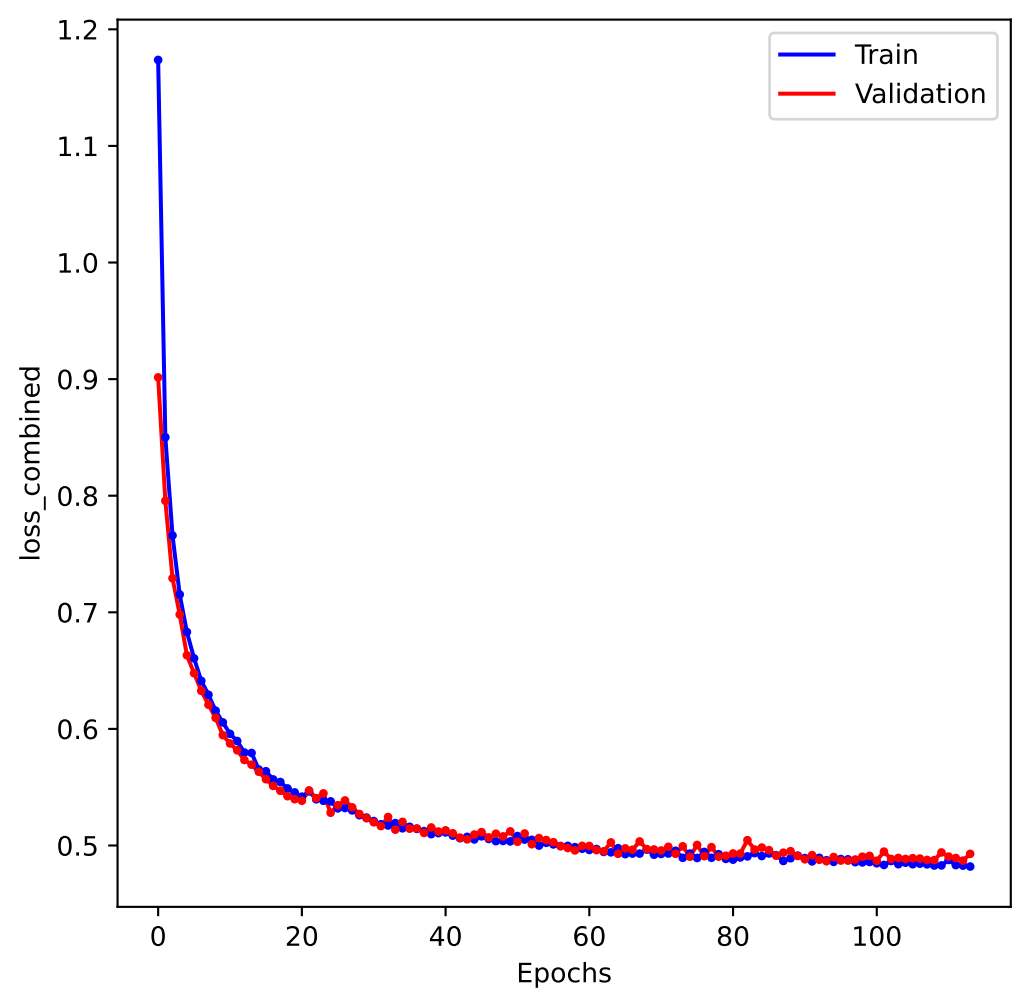

Test estimates: exp_scalar_w
MAE: 1.27E-01
MSE: 3.18E-02
RMSE: 1.78E-01
Intercept: 1.68E-01
Slope: 5.39E-01
Coverage: 79.2%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
0.2
0.4
0.6
0.8

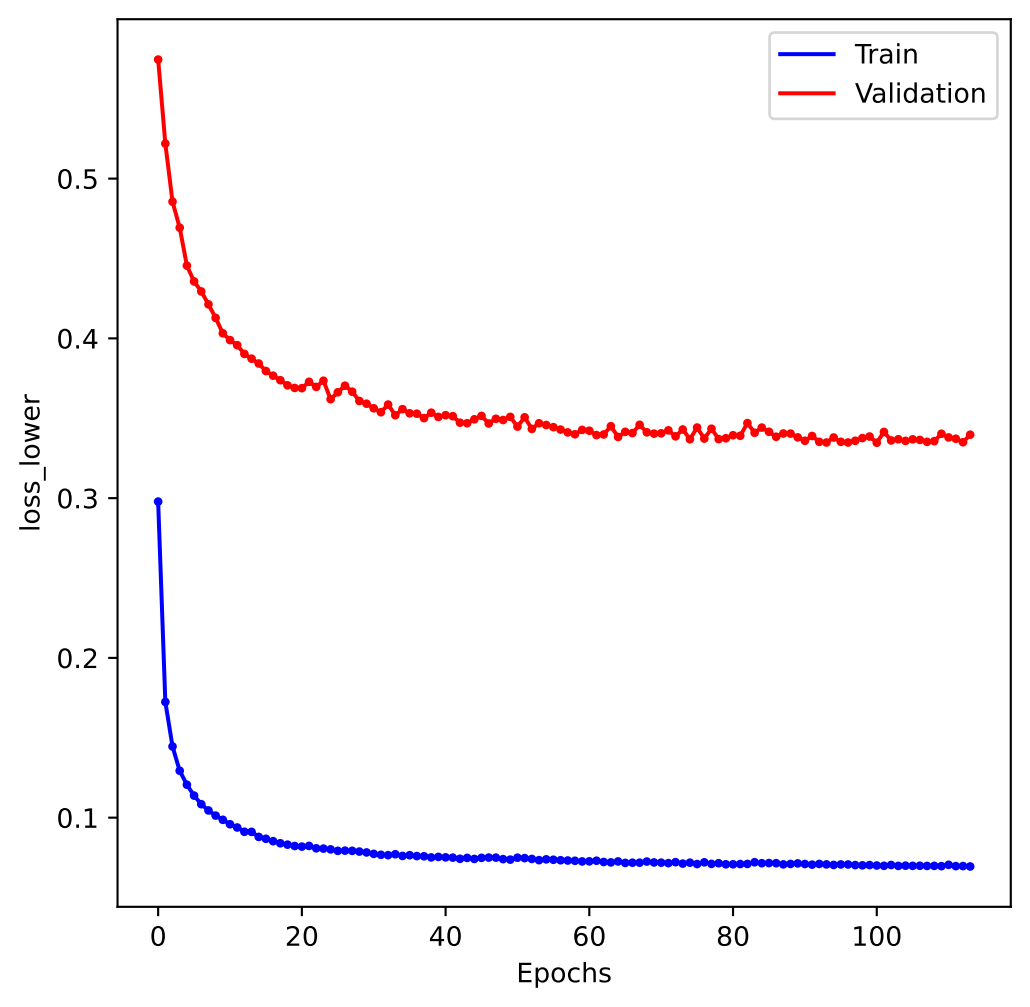

Test estimates: log_base_b
MAE: 7.56E-01
MSE: 1.73E+00
RMSE: 1.32E+00
Intercept: -5.42E-01
Slope: 6.57E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

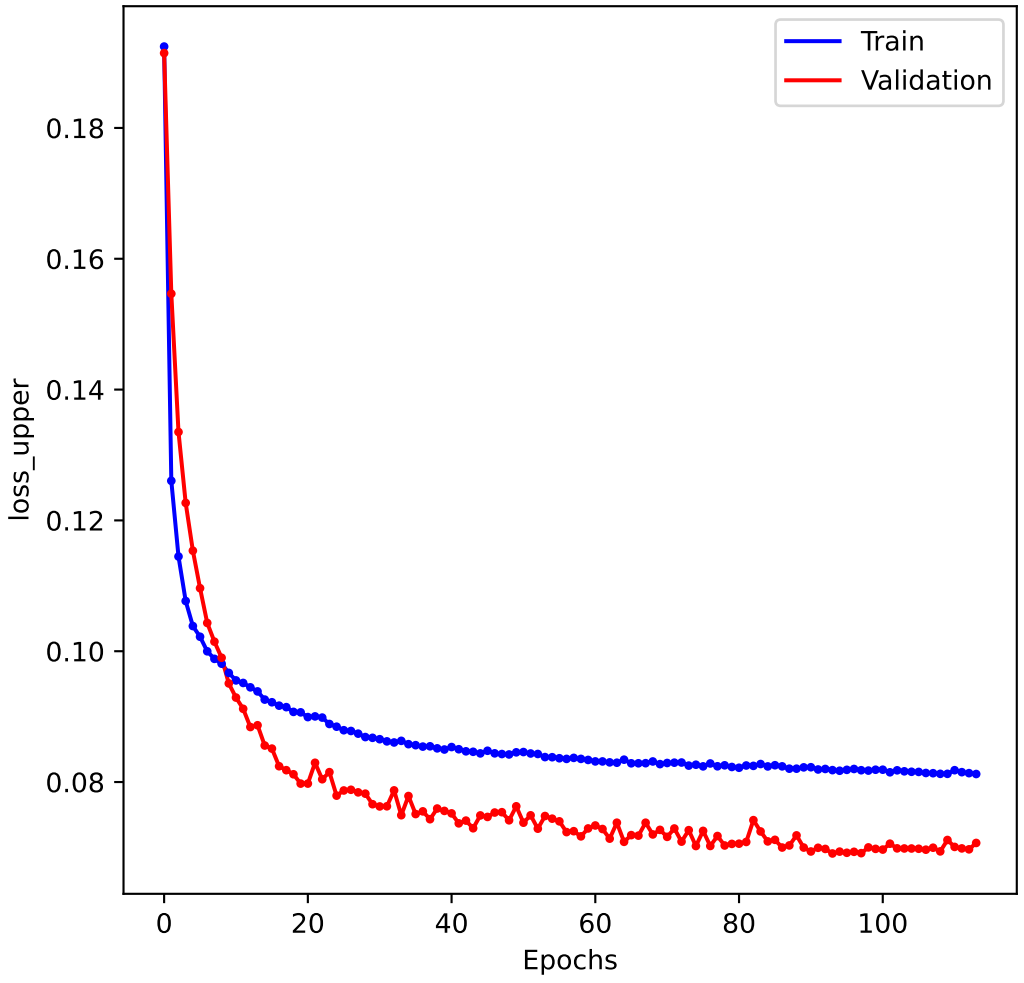

Test estimates: log_base_d
MAE: 4.91E-01
MSE: 4.56E-01
RMSE: 6.75E-01
Intercept: -4.05E-02
Slope: 8.90E-01
Coverage: 79.6%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

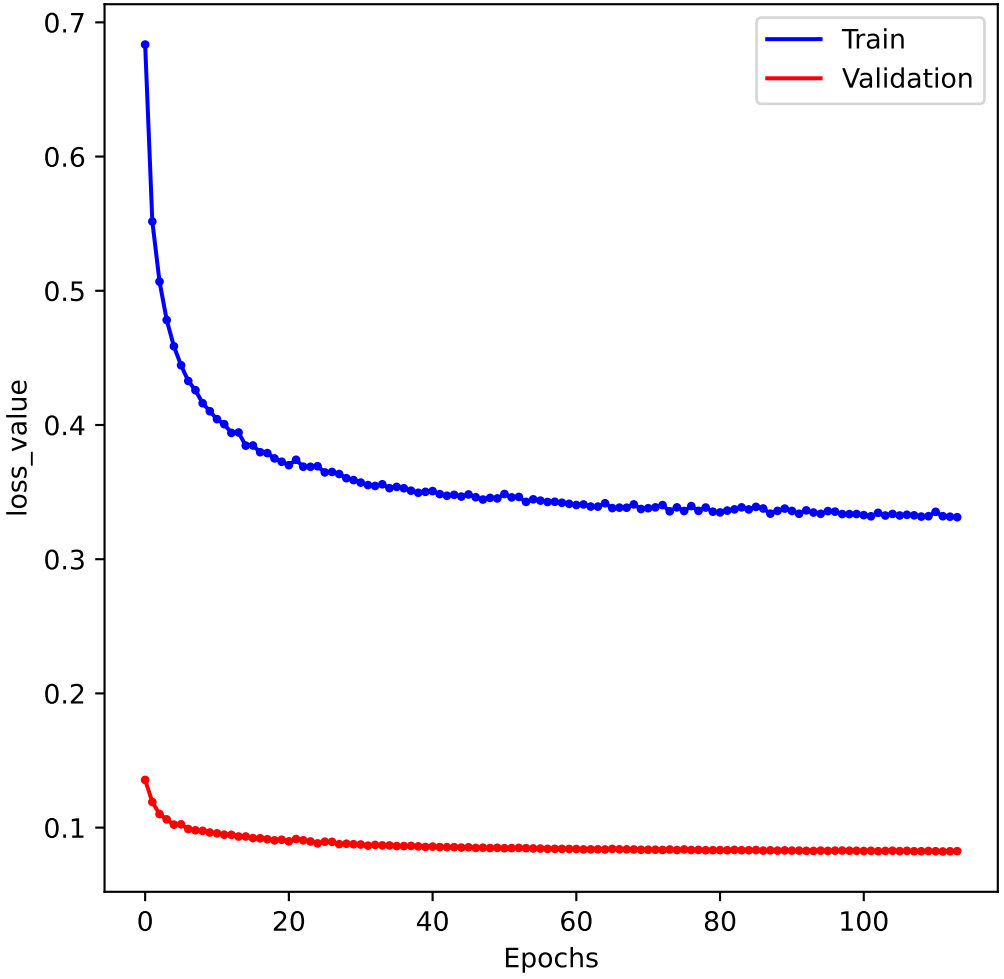

Test estimates: log_base_e
MAE: 4.24E-01
MSE: 5.17E-01
RMSE: 7.19E-01
Intercept: -2.09E-01
Slope: 8.99E-01
Coverage: 80.4%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

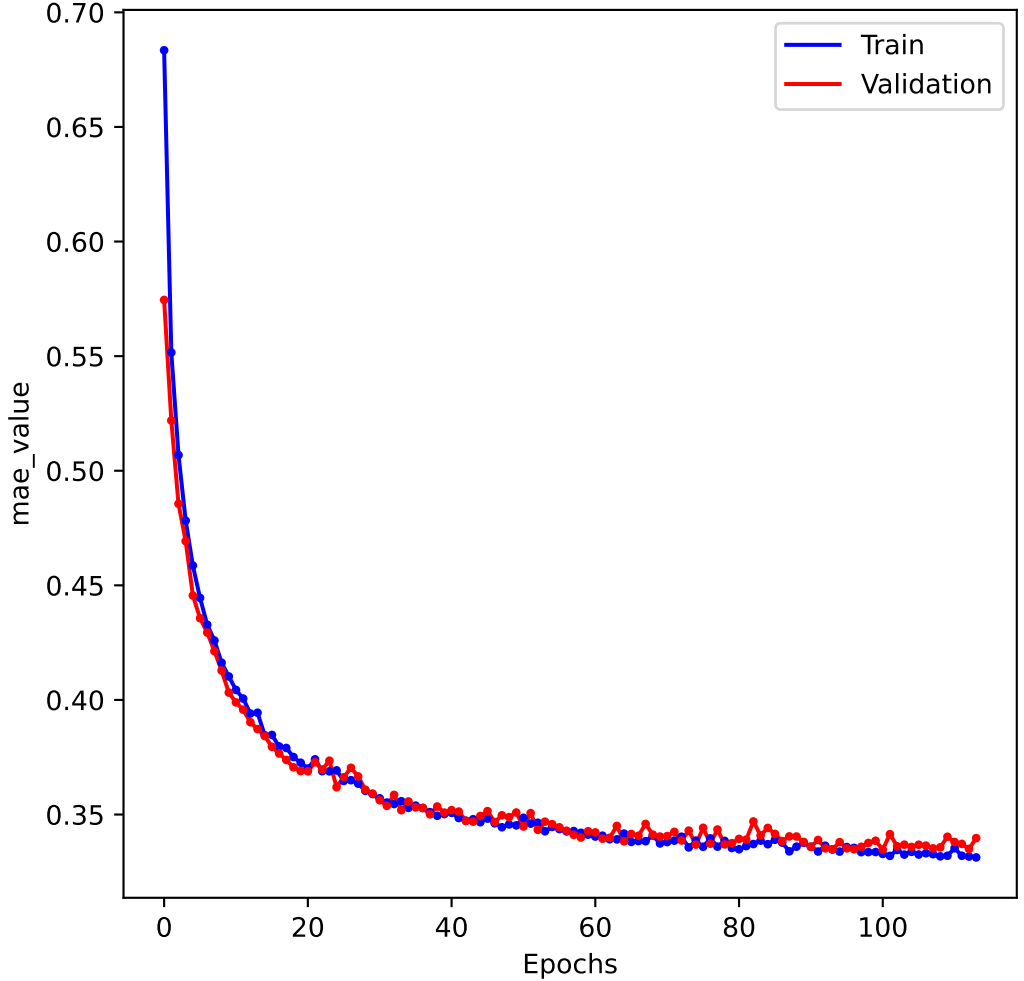

Test estimates: log_base_w
MAE: 6.14E-01
MSE: 9.09E-01
RMSE: 9.54E-01
Intercept: -3.82E-01
Slope: 7.59E-01
Coverage: 79.2%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8

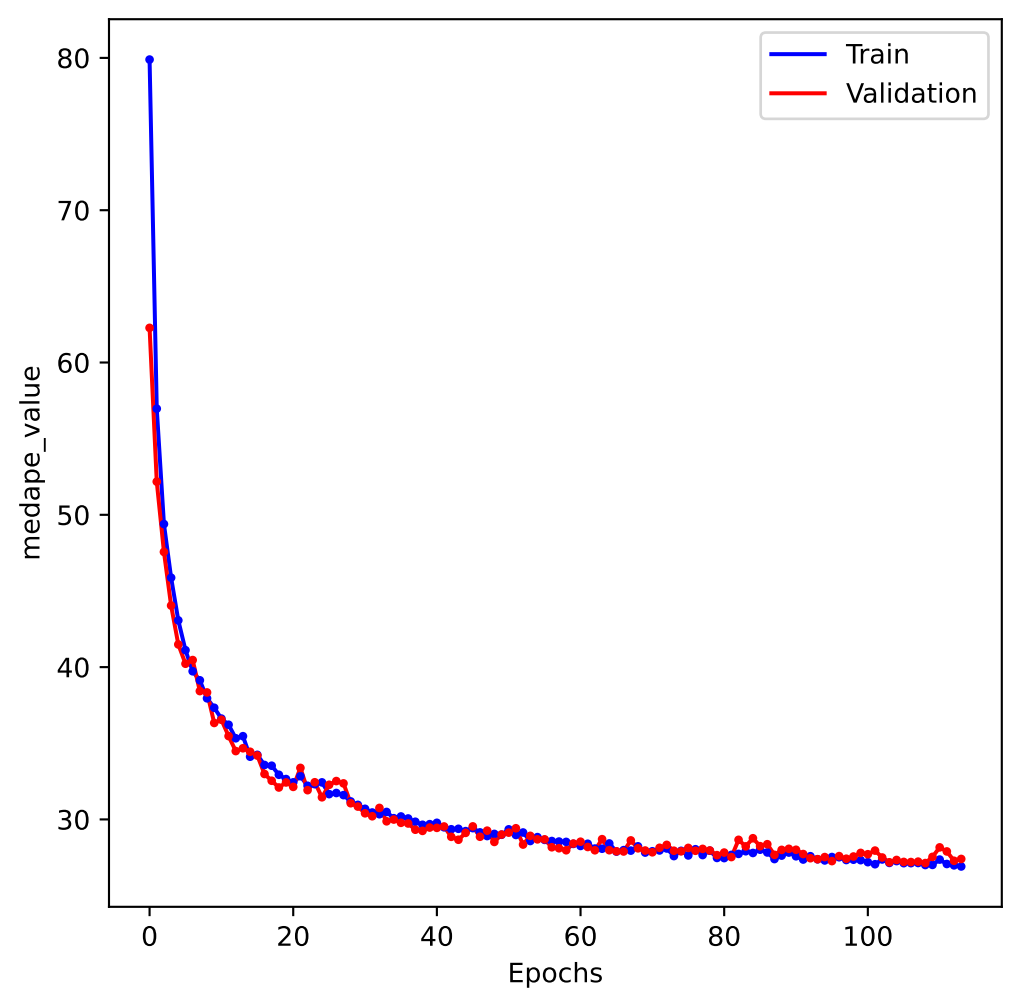

Train
Validation
loss_combined
Epochs
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0
20
40
60
80
100

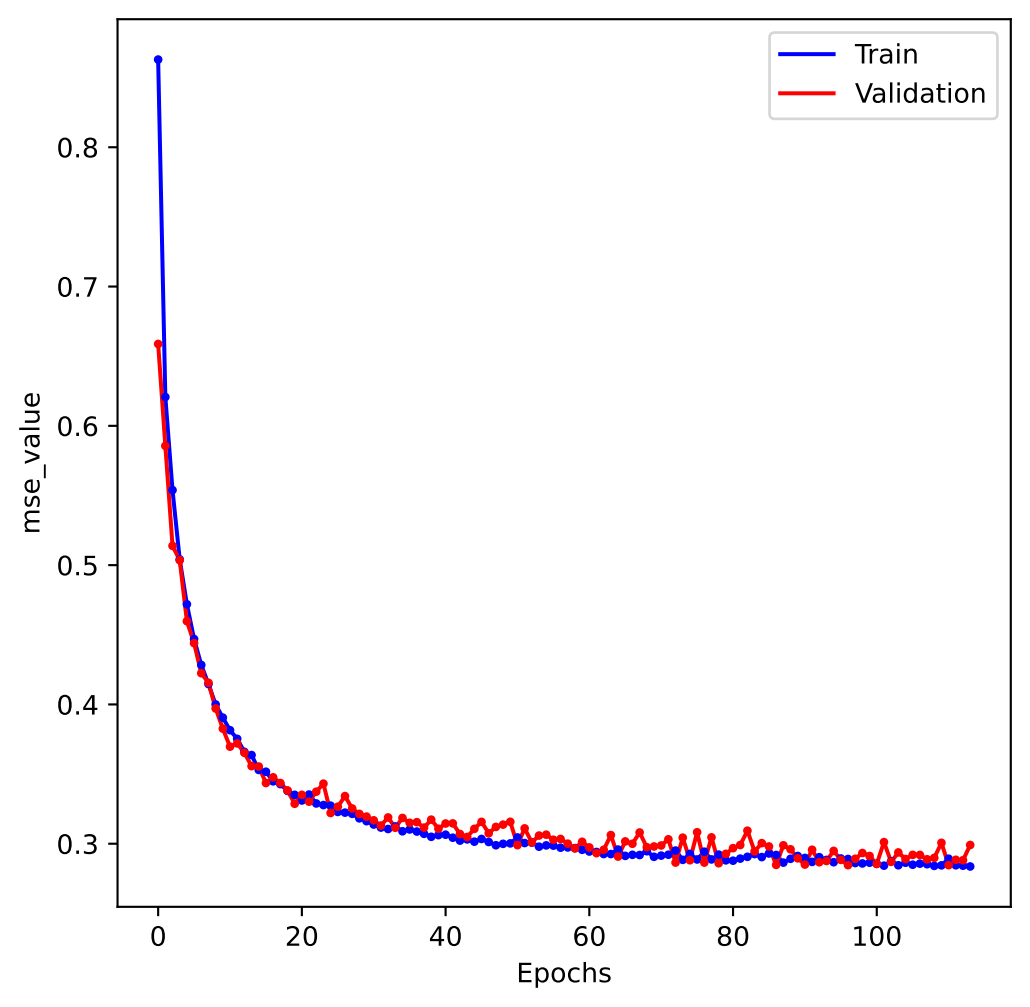

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
Epochs

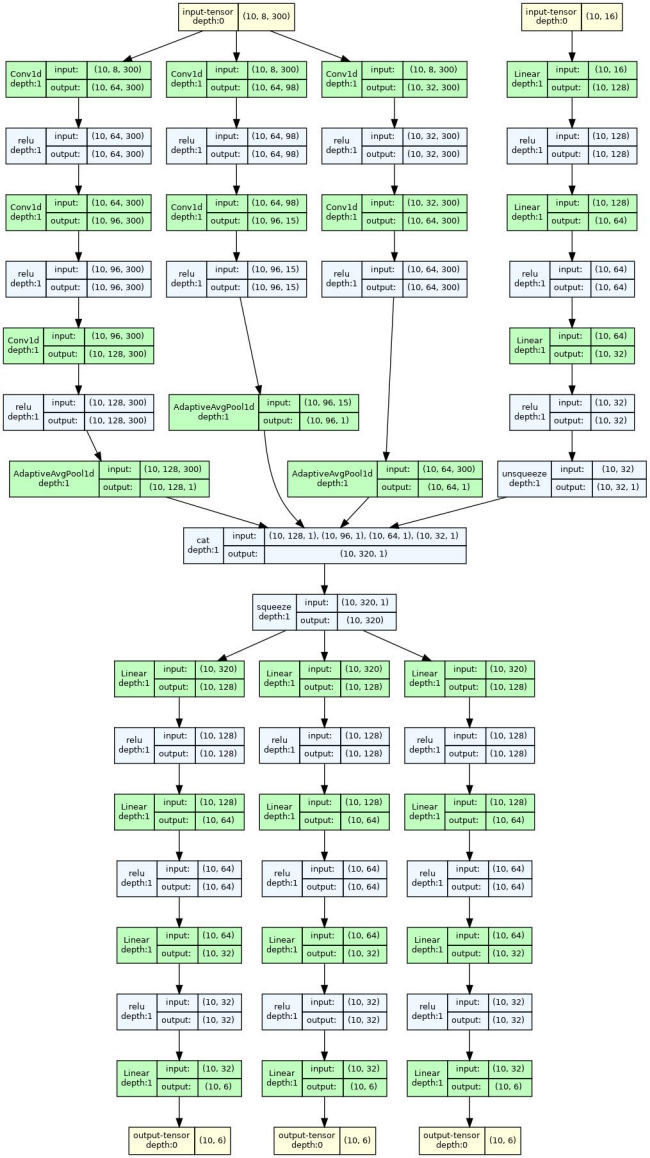

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0
20
40
60
80
100
Epochs


Train
Validation
loss_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100


Train
Validation
mae_value
Epochs
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0
20
40
60
80
100


Train
Validation
medape_value
Epochs
80
70
60
50
40
30
0
20
40
60
80
100


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0
20
40
60
80
100
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)